Nguyen Ngoc Gia Ninh *(Aaron)*

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Research interests: computer vision for marine and environmental monitoring; data-centric and label-efficient learning for underwater imagery; building expert-in-the-loop datasets and annotation systems.

Education

Bachelor in Information Technology, Artificial Intelligence specialisation — Sep 2021 – Mar 2025
FPT University — Ho Chi Minh City, Vietnam

- Degree conferred March 2025 (commencement November 2025). Cumulative GPA 7.78/10 (3.25/4.0) over 145 credits, classification *Good*. Programme taught in English.
- Capstone thesis: *Applied AI for Healthy – An Automatic Lip Reading System based on Deep Learning* (grade 8.2/10).
- Honor Student, Summer Semester 2023.

Publications and Research Outputs

Order as listed by the publisher or repository.

[M1] **N. N. G. Ninh**, et al. *NTCoral: an expert-annotated image dataset of Vietnamese coral reefs* (working title). Dataset paper in preparation, first author.

[B1] N. P. T. Thao, **N. N. G. Ninh**, T. N. Quoc, V. T. Hoang. "A Multistage Deep Learning Approach for Intracranial Aneurysm Detection." In *Intelligent Precision Healthcare*, CRC Press, pp. 287–301, 2026.

[P1] Q.-H. N. Dam, P. H. Nguyen, M. P. Tran, C.-S. Trinh, T. T. Nguyen, **N. N. G. Ninh**, et al. "From Monitoring to Action: A Comprehensive Survey of Autonomous Decision-Making in Smart Aquaculture Systems." Preprint, 2026; under review at *Computers and Electronics in Agriculture*.

[P2] T. N. Quoc, N. P. T. Thao, **N. N. G. Ninh**, et al. "A Large-Scale Vietnamese Lip Reading Dataset and Enhanced Recognition Pipeline." Preprint, 2026.

[D1] T. N. Quoc, N. P. T. Thao, **N. N. G. Ninh**, H. D. N. Quynh, V. T. Hoang, et al. *A Large-Scale Vietnamese Lip-Reading Dataset for Visual Speech Recognition*. Zenodo, v3, 2026. CC BY 4.0.

Research and Professional Experience

Research Assistant — Apr 2026 – present
Smart Green Transformation Center (GREEN-X), VinUniversity; supervisor: Phung Manh Duong — Hanoi, Vietnam

- Collaborating with the group since December 2025; formal research-assistant contract from April 2026.
- Lead the NTCoral project: an expert-verified image dataset of Vietnamese coral reefs built from underwater video surveys, with traceable links from every image to its source video and frame.
- Coordinate video review and genus-level annotation with marine biology experts; the first expert-labelled set covers 200 images and 1,679 annotated regions.
- Design the frame-selection and image-quality pipeline (spatial deduplication, quality screening, content coverage) and evaluate few-shot segmentation models (DINOv3 + SAM2) on local survey data.
- Authored Section 6, *Autonomous Decision-Making and Closed-Loop Control*, of the smart-aquaculture survey [P1].

Technology Specialist (AI research track) — Jul 2025 – Apr 2026
VinBigData — Hanoi, Vietnam

- Selected for an intensive research and training programme covering mathematics, computer vision, NLP, generative AI, data science and MLOps.

AI Researcher — Jan 2024 – Jul 2025
Perception Technology Solutions — Remote

- Vietnamese visual speech recognition: sourced and pre-processed video for the VLR lip-reading dataset (dataset; clip segmentation, alignment, mouth-region extraction) and trained and evaluated VSP-LLM and Auto-AVSR baselines [P2, D1].
- Deployed lightweight lip-reading models on an NVIDIA Jetson Nano for real-time inference experiments.
- VNPlant-255 medicinal plant dataset: pre-processing, annotation and benchmarking of image classification architectures (PyTorch, timm).
- Traffic-video analysis: vehicle detection and multi-object tracking components.

AI Engineer Intern Sep 2023 – Dec 2023
FPT Software AI Valley Quy Nhon, Vietnam

- Compared PhoBERT and XLM-R for Vietnamese review analysis (aspect classification and rating prediction); served the model through a FastAPI service in Docker with a browser demo (code).

Teaching Assistant Nov 2024 – May 2025
TEKY Academy of Technology Ho Chi Minh City, Vietnam

- Supported primary-school students in programming exercises, debugging and project-based coding.

Certificates and Languages

- TensorFlow Developer Certificate, Google, March 2024 (valid to March 2027).
- Vietnamese (native); English (professional working proficiency; undergraduate programme taught in English).

References

Available upon request.